# Math 2551 Worksheet: Chain Rule

1. An object travels along a path on a surface. The exact path and surface are not known, but at time $t = t_0$ it is known that

$$\frac{\partial z}{\partial x} = 5, \quad \frac{\partial z}{\partial y} = -2, \quad \frac{dx}{dt} = 3, \quad \frac{dy}{dt} = 7.$$

   Use this information to determine the rate of change of the height $z$ of the object with respect to time $t$ at $t = t_0$.

2. Find the values of $t$ where $\frac{dz}{dt} = 0$ if $z = 3x + 4y$, $x = t^2$, and $y = 2t$.

3. Let $w = xy + yz + zx$, where $x = r\cos\theta, \quad y = r\sin\theta, \quad z = r\theta$. Find $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial \theta}$ when $r = 2$ and $\theta = \frac{\pi}{2}$.

4. Suppose we have a differentiable function $w = g(x, y)$ and $x$ and $y$ are differentiable functions of $t$ and we know the following information.

$$g(1,0) = 1,\ g_x(1,0) = -2,\ g_y(1,0) = 2,\ g(-1,2) = 3,\ g_x(-1,2) = 1,\ g_y(-1,2) = -2,$$

$$x(2) = 1,\ y(2) = 0,\ x(1) = 1,\ y(1) = 3,\ x'(2) = 4,\ y'(2) = -1,\ x'(1) = 0,\ y'(1) = 2$$

   If possible, find $\frac{dw}{dt}(1)$ and $\frac{dw}{dt}(2)$ or explain why the given information is not enough to do so. Which of these pieces of information would you not use at all to compute either value?